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Cultural Influences on Personality Types: A Cluster Analysis of HEXACO Traits in Japan

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ABSTRACT This study investigates the structure of personality types among Japanese individuals using the HEXACO model, based on data from 13,668 participants. Although personality trait research has made significant progress, the existence and consistency of personality types remain under debate. To address this, we applied a combination of Ward's hierarchical clustering and k-means analysis, supplemented by latent profile and spectral clustering to ensure robustness. The results revealed seven distinct personality types that capture both universal and culture-specific dimensions of personality. These types include Reserved, Emotionally Sensitive, Conscientious Introvert, ~~Extraverted Agreeable,~~Exploratory Extravert, Self-Oriented Independent, ~~Extraverted Stable~~Emotionally Volatile Extravert, and ~~Introverted Leader~~Reliable Introvert. The Honesty–Humility factor, central to the HEXACO framework, was particularly influential in explaining cultural aspects of Japanese personality, reflecting the balance between humility, emotional restraint, and social harmony emphasized in Japanese society. Moreover, the study demonstrated how methodological diversity and large-scale data enhance the reliability of typological research in non-Western contexts. The findings extend previous Western-centered studies by providing empirical evidence of distinct personality configurations shaped by Japanese cultural values. This study contributes to cross-cultural psychology by highlighting the universality and specificity of HEXACO-based personality structures and underscores the need for culturally grounded models in global personality science.

INDEX TERMS cluster analysis, cultural psychology, HEXACO model, non-WEIRD, personality types

I. INTRODUCTION

~~Personality research seeks to explain the consistent patterns of thoughts, emotions, and behaviors that distinguish individuals. Over the past few decades, advances in psychometrics and data science have expanded the field beyond trait-level analysis to include cross-cultural and typological approaches. However, most studies have been conducted in Western populations, leaving a gap in understanding personality structures in East Asian contexts. Japan, in particular, presents a unique cultural environment emphasizing humility, harmony, and emotional sensitivity, which may shape personality configurations in distinctive ways.~~

~~This study applies the HEXACO model—a six-dimensional framework including Honesty–Humility—to explore personality types within a large Japanese sample. By combining multiple clustering techniques and validity indices, this research aims to identify robust personality types and examine how cultural factors influence their formation. The findings are expected to contribute to both theoretical refinement of the HEXACO model and practical applications in cultural and organizational psychology. The following sections present a detailed review of the theoretical background, describe the analytical procedure, and discuss the implications of the identified personality types for understanding cultural variations in personality structure.~~Personality traits play a central role in understanding how individuals perceive, interpret, and adapt to their environments. Over the past decades, a range of theoretical models has been proposed to explain the multidimensional structure of personality. Among these, Cloninger's psychobiological model has been particularly influential for its integration of biological, developmental, and psychological perspectives. The model distinguishes between temperament—biologically rooted tendencies such as novelty seeking, harm avoidance, reward dependence, and persistence—and character, which reflects self-directedness, cooperativeness, and self-transcendence

shaped by learning and socialization processes [1]. This framework emphasizes the dynamic interplay between innate predispositions and life experiences, offering a comprehensive account of personality development and variability across the lifespan.

Importantly, psychobiological traits have been shown to exhibit systematic associations with psychopathology. A recent meta-analysis of 149 studies demonstrated robust and transdiagnostic links between temperament and character dimensions and a wide range of mental disorders [2]. High harm avoidance and low self-directedness emerged as broad vulnerability markers across more than 20 diagnostic categories, while novelty seeking, reward dependence, and self-transcendence also displayed consistent associations with mood, substance use, and psychotic-spectrum conditions. These findings position personality structure not merely as a descriptive taxonomy of individual differences but as a framework deeply intertwined with psychological adjustment and clinical risk. This broader theoretical landscape reinforces the importance of examining culturally shaped personality configurations, as cultural norms may moderate both trait expression and their associations with mental health.

Beyond biologically informed models, trait-based frameworks such as the HEXACO model have gained widespread attention for their cross-cultural applicability and conceptual expansion of the Big Five. The HEXACO model introduces Honesty–Humility as a sixth core dimension, capturing fairness, sincerity, and prosocial tendencies—domains with well-established evolutionary and cultural significance. Previous studies across North American and European populations show that the six-factor structure replicates reliably across diverse languages and contexts, and offers strong predictive validity for prosocial behavior, workplace functioning, moral decision-making, and interpersonal outcomes. Moreover, recent work highlights meaningful cross-cultural differences in mean levels of Honesty–Humility and Emotionality, reflecting cultural norms concerning modesty, social harmony, emotion regulation, and interpersonal sensitivity [3], [4]. These findings illustrate that personality expression cannot be fully understood without considering cultural factors that shape both trait manifestation and social expectations.

Despite intensive research at the dimensional level, fewer studies have examined personality prototypes—naturally occurring configurations of multiple traits that may better reflect how individuals function in daily life. Prototype-based approaches complement dimensional perspectives by capturing the interactional patterns among traits, offering intuitive frameworks applicable to clinical assessment, organizational contexts, and behavioral prediction. Prior research, largely grounded in Western populations, has consistently identified prototypes such as the "Resilient," "Undercontrolled," and "Overcontrolled" types, though additional clusters appear depending on sample characteristics and analytic techniques. Notably, prototype structures often vary across cultural settings, suggesting that trait configurations emerge from an interplay of universal psychological processes and sociocultural norms.

In East Asian societies, cultural constructs such as collectivism, interpersonal harmony, modesty, and emotional restraint may systematically influence the co-occurrence of HEXACO traits. For example, Japanese samples frequently show lower Extraversion and higher Emotionality compared to Western populations, while longstanding cultural norms surrounding modesty and self-effacement may shape the expression of Honesty–Humility. These tendencies raise the possibility that personality prototypes derived from Western samples may not generalize to East Asian populations. Yet, to date, no large-scale prototype analysis of HEXACO traits has been conducted in Japan, leaving a significant gap in the literature on culturally grounded personality structure.

The present study addresses this gap by applying multiple clustering techniques to a large Japanese adult sample (N = 13,668) to identify robust, data-driven personality prototypes based on the HEXACO framework. This investigation contributes to the literature in several key ways.

First, it represents the largest personality prototype analysis conducted in a non-WEIRD East Asian population using HEXACO traits.

Second, it employs a multi-method clustering framework—including Ward's method combined with k-means, latent profile analysis, and spectral clustering—to enhance replicability and methodological rigor.

Third, it offers a culturally informed interpretation of the resulting prototypes, extending Western-centric findings and illuminating how cultural norms may shape patterns of trait covariation.

By integrating perspectives from psychobiology, clinical personality research, and cultural psychology, this study advances our understanding of how personality structure emerges within a distinctive sociocultural context. The findings also provide a foundation for future cross-cultural comparisons, applied psychological assessments, and investigations into culturally moderated personality–behavior associations.

II. BACKGROUND AND HISTORY

Personality research is one of the longest-standing fields in psychology, with efforts to understand individual differences in behavior and personality dating back to ancient Greece [~~1~~5]. Modern personality research has largely focused on the Five-Factor Model (FFM), established between the 1980s and 1990s [~~2~~],[~~36~~],[7]. This model seeks to describe individual personality using five major dimensions: Extraversion, Agreeableness,

Conscientiousness, Neuroticism, and Openness to Experience.

Since the establishment of the FFM, personality research has expanded to integrate insights from genetics, neuroscience, and evolutionary psychology, aiming to uncover the biological basis of personality traits. Twin studies have indicated that the heritability of personality traits ranges from 40% to 60% [48], while neuroimaging studies have linked personality traits to specific brain regions and neural circuits [~~5~~9]. From an evolutionary psychology perspective, these traits are hypothesized to have played adaptive roles in human evolution [~~6~~10]. Personality traits are associated with a wide range of psychological constructs, including well-being and life satisfaction [~~7~~11], psychopathology [~~8~~12], and cognitive abilities [~~9~~13]. These findings suggest that personality plays a crucial role in psychological adaptation and functioning. Moreover, personality traits have been found to influence various aspects of life. In the realm of academic achievement, Conscientiousness has shown the strongest positive correlation, followed by Openness [~~10~~14]. In career-related domains, traits such as Extraversion have been associated with higher compatibility in sales and leadership roles [~~11~~15]. In interpersonal relationships, Agreeableness has been linked to the formation and maintenance of positive social relationships [~~12~~16]. Regarding health, higher levels of Neuroticism have been identified as risk factors for various physical and mental health issues [~~13~~17]. Thus, personality research has not only contributed to our understanding of individual differences but has also advanced knowledge of their biological underpinnings, environmental interactions, and their influence across multiple life domains. Future directions include the development of more refined measurement techniques, the adoption of longitudinal study designs, and the promotion of cross-cultural comparative research.

The HEXACO Model

The HEXACO model is a relatively new personality model proposed by Ashton and Lee in the early 2000s [~~14~~18]. This model expands upon the traditional Big Five model, including six major dimensions: Honesty-Humility, Emotionality, Extraversion, Agreeableness, Conscientiousness, and Openness to Experience. The acronym "HEXACO" is derived from the initials of these six factors. One of the most distinctive features of the HEXACO model is the addition of the Honesty-Humility factor, which includes traits related to sincerity, fairness, modesty, and low greed. By incorporating this factor, the HEXACO model allows for a clearer description of moral and ethical aspects of personality, which were not adequately captured by the Big Five [~~15~~19]. A key difference between the HEXACO and Big Five models lies in the reconceptualization of Agreeableness and Neuroticism (Emotionality in the HEXACO model). HEXACO's Agreeableness focuses on traits such as tolerance and flexibility, characterized by the absence of anger, while Emotionality includes fearfulness, anxiety, dependence, and sentimentality, differentiating it somewhat from the Big Five's Neuroticism [~~16~~20]. The HEXACO model has been shown to outperform the Big Five in explaining traits associated with the Dark Triad (Machiavellianism, psychopathy, and narcissism). For example, the Honesty-Humility factor has demonstrated a strong negative correlation with these dark traits [~~17~~21]. Furthermore, the HEXACO model has exhibited superior predictive power in relation to socially significant behaviors and attitudes, such as counterproductive workplace behavior [~~18~~22], risk-taking [~~19~~23], and substance use [~~20~~24]. The validity of the HEXACO model has also been supported by studies across different languages and cultures. Ashton et al. [~~21~~25] analyzed lexical studies in 12 different languages, consistently identifying the six-factor structure of the HEXACO model, suggesting that it reflects a universal personality structure across cultures. In addition, the HEXACO model is supported by evolutionary psychology. In particular, the Honesty-Humility factor is thought to be associated with the evolution of cooperative behavior in humans [~~14~~18]. Individuals high in Honesty-Humility tend to prioritize long-term reciprocal cooperation over short-term self-interest, which is seen as an adaptive trait in human societies. The HEXACO model brings new perspectives to personality research, enabling a more comprehensive understanding of human behavior and attitudes. Future research challenges include elucidating the neurobiological foundations of the model, exploring developmental changes through longitudinal studies, and expanding cross-cultural comparisons. Further investigation into the utility of the HEXACO model in applied fields such as clinical and organizational psychology is also warranted.

Current Research on Personality Types

Understanding personality types is essential for capturing the complexity of human individuality beyond isolated traits. While variable-centered approaches offer valuable insights into the role of specific personality dimensions, person-centered approaches allow for a more holistic understanding by identifying recurring constellations of traits that reflect distinct personality types [~~22~~26]. This approach not only enhances psychological theory by bridging trait and typological perspectives, but also has practical implications for fields such as clinical, organizational, and educational psychology, where recognizing consistent personality patterns can inform targeted interventions and support strategies [~~23~~],[~~24~~27],[28]. Moreover, as personality types are shaped by both universal and culture-specific influences, investigating them across diverse populations—particularly using models like HEXACO that account for moral dimensions such as Honesty-Humility—can deepen our understanding of the interaction between personality and cultural context [~~14~~18]. Given the current lack of large-scale personality type studies in non-Western settings, especially in East Asia, advancing this line of research is

crucial for developing a more globally inclusive personality science.

In personality research, two main methodological approaches exist: the variable-centered approach and the person-centered approach. The variable-centered approach treats individual personality traits independently, focusing on their continuous variation. In contrast, the person-centered approach seeks to identify patterns or types of individuals characterized by specific combinations of multiple traits [~~22~~26]. Research on personality types follows the latter approach, aiming to understand individuals holistically.

The history of personality type research dates back to the ancient Greek theory of the four humors, but its modern foundations were laid by Block and Block's [~~25~~29] theory of ego control and ego resiliency. Based on this theory, Robins et al. [~~26~~30] proposed three basic personality types: resilient, overcontrolled, and undercontrolled. These types have been replicated in many subsequent studies, becoming known as the ARC types (Asendorpf, Robins, Caspi). Resilient individuals exhibit low Neuroticism and high levels of other traits, reflecting adaptability to the environment. Overcontrolled individuals are characterized by high Neuroticism and low Extraversion, showing strong inhibition of emotions and impulses. Undercontrolled individuals, on the other hand, display high Extraversion and low Agreeableness and Conscientiousness, showing impulsive and less rule-bound behavior. Recent research has re-examined personality types using larger datasets and more advanced statistical methods. Daljeet et al. [~~23~~27], using latent profile analysis with the HEXACO model, identified five personality types: socially considerate, confident, goal-oriented, withdrawn, and maladaptive. Espinoza et al. [~~24~~28], in a study involving over 270,000 participants, also identified five optimal personality profiles: egocentric, flexible, disciplined, sociable, and role model types. Sava and Popa [~~27~~31], conducting a cluster analysis on a Romanian sample, identified five personality types, including resilient, overcontrolled, and undercontrolled types, as well as average and reserved types. In Japan, Kase et al. [~~28~~32], based on Big Five traits, identified four personality types, suggesting the existence of a "Not Identifiables" type unique to Japan, characterized by low Extraversion and high Agreeableness. ~~Gerlach et al. [29~~In addition to these earlier findings, recent work has applied latent profile analysis to integrate HEXACO traits with socially aversive dimensions such as the Dark Triad. She et al. (2025) identified five robust personality profiles across UK and Dutch samples and demonstrated that profile structures are partly universal but also shaped by cultural norms. Their results highlight how person-centered models can uncover non-linear combinations of traits that differ from traditional trait-level interpretations, underscoring the importance of examining culturally grounded personality configurations in non-Western contexts [33]. Gerlach et al. [34] reported four distinct personality types in a study involving over 1.5 million participants: average, role model, egocentric, and reserved types. Kerber et al. [~~30~~35], reviewing the literature and using a large German dataset, concluded that five personality types (resilient, overcontrolled, undercontrolled, reserved, and vulnerable-resilient) were optimal. These existing studies highlight several challenges. First, the choice of clustering algorithms can significantly impact results. Many studies use only a single algorithm, but comparing multiple algorithms can yield more robust results. Second, the methods for evaluating cluster solution validity remain a concern. It is important to consider both internal and external validity and use multiple indicators for comprehensive evaluation. Third, much of the existing research is based on Western cultural data, raising the possibility that cultural diversity is underrepresented. Cross-cultural comparisons are needed to clarify the universality and specificity of personality types. Finally, studies using the HEXACO model for personality type research are still relatively rare, particularly in non-Western cultures. Further research is needed to investigate how the HEXACO model's Honesty-Humility factor contributes to the formation of personality types. Addressing these challenges can help establish a stronger scientific foundation for personality type research and contribute to the advancement of individual differences psychology.

The Relationship Between Culture and Personality

The relationship between culture and personality is an important and complex theme in psychological research. The universality hypothesis of personality traits posits that the basic structure of personality is common across cultures [~~3~~7]. This hypothesis is supported by findings that major personality models, such as the Big Five and HEXACO, are consistently replicated across different cultural contexts [~~14~~18]. However, while the universality of personality structures is supported, cultural differences cannot be ignored. Significant differences in the expression and evaluation of personality traits are observed between Western and Eastern cultures. For example, Western cultures tend to emphasize individualistic values, where self-assertion and individual expression are prioritized, whereas Eastern cultures, including Japan, tend to emphasize collectivist values, where relationships with others and harmony are prioritized [~~31~~36]. These cultural differences may influence the manifestation and interpretation of personality traits. In the context of Japanese culture, several distinctive trends have been reported. Schmitt et al. [~~32~~37], in a large-scale cross-cultural study involving 56 countries, found that compared to other nations, Japanese individuals exhibited the following traits:

- High Neuroticism: Japanese individuals tend to experience negative emotions such as anxiety and depression more frequently.
- Low Conscientiousness: Japanese individuals scored lower in aspects of planning and self-discipline compared to other nations.

- Low Extraversion: Japanese individuals tend to be more reserved and less sociable.
- Low Agreeableness: Japanese individuals scored slightly lower in cooperation and altruism.
- Low Openness: Japanese individuals tended to be more conservative and less open to new experiences.

These traits are closely related to Japanese cultural values and social norms. Notably, the high level of Neuroticism has been confirmed in studies by Kase et al. [~~28~~32], who suggested the existence of a "Not Identifiables" type unique to Japan, characterized by low Extraversion and high Agreeableness. Furthermore, Jonason et al. [~~33~~38] found that Japan exhibited higher tendencies of Machiavellianism and psychopathy, two of the Dark Triad traits, compared to other countries. While this may seem contradictory to traditional stereotypes of Japanese culture, these findings may reflect the complex social environment in Japan, where competitive pressures and the pursuit of self-interest, while maintaining outward harmony, play a role in the elevated presence of these traits. Personality research based on the HEXACO model in Japan is still limited, but the Honesty-Humility factor may offer particularly intriguing insights within the context of Japanese culture. The contrast between superficial humility and the relatively higher levels of Machiavellianism in actual behavior may provide new insights into this factor's role. However, the accumulation of HEXACO data on Japanese individuals remains insufficient. To further explore the relationship between culture and personality, it is essential to conduct additional large-scale studies using the HEXACO model in Japan. Understanding the relationship between culture and personality is vital in today's increasingly globalized society. To promote mutual understanding and effective communication with people from different cultural backgrounds, it is necessary to consider both the universal and culture-specific aspects of personality. Future research should aim to unravel the complex interaction between culture and personality using more sophisticated measurement methods and analytical techniques.

Purpose and Significance of the Present Study

The primary aim of this study is to identify personality types using a large dataset of Japanese individuals based on the HEXACO model and to clarify the characteristics of these types. To achieve this goal, we established the following three research questions:

Research Question 1: Do the number and characteristics of personality types derived from the Japanese HEXACO data align with those reported in previous studies? This research question seeks to examine the extent to which the personality structure of Japanese individuals aligns with the results of previous studies centered around Western cultures. Studies such as those by Daljeet et al. [~~23~~27] and Espinoza et al. [~~24~~28] have identified five personality types based on the HEXACO model. In this study, we will investigate whether the number of types extracted from the Japanese data matches the results of prior studies and whether the characteristic profiles of each type are similar. This comparison is expected to provide new insights into the universality and cultural specificity of personality types.

Research Question 2: Do the characteristics of the extracted personality types reflect the cultural characteristics of Japan? This question explores how the personality types of Japanese individuals reflect unique features of Japanese culture. According to Schmitt et al. [~~32~~36], Japanese individuals exhibit higher levels of Neuroticism and lower levels of Conscientiousness, Extraversion, Agreeableness, and Openness compared to other nations. Kase et al. [~~28~~32] also suggested the existence of a unique "Not Identifiables" type in Japan. Furthermore, Jonason et al. [~~33~~37] reported higher tendencies of Machiavellianism and psychopathy among Japanese individuals. We will analyze how the personality types extracted in this study reflect these cultural tendencies. This analysis is expected to deepen our understanding of the interaction between culture and personality.

Research Question 3: How do the six factors of the HEXACO model contribute to the formation of personality types in Japanese individuals? This research question examines the role of the six factors of the HEXACO model, particularly the Honesty-Humility factor, in shaping the personality types of Japanese individuals. As a unique factor not included in the Big Five model, the Honesty-Humility factor may play a critical role in understanding the emphasis on humility in Japanese culture, as well as the apparent contradiction between superficial humility and high levels of Machiavellianism. By analyzing the contribution of each factor, we aim to gain a more detailed understanding of the personality structure of Japanese individuals.

III. MATERIALS AND METHODS

A. STUDY DESIGN AND PARTICIPANTS

This study employed a cross-sectional online survey design. Participants were recruited through public posts on social networking platforms (SNS) in Japan. Eligibility criteria required that individuals be (a) residents of Japan, (b) at least 18 years of age, (c) proficient in Japanese, and (d) able to complete all items of the personality questionnaire. Responses were collected between July 1 and September 30, 2024, using Google Forms. To prevent multiple submissions, each participant was required to log in with a Google account, which restricted participation to a single response per account.

A total of 13,668 participants were included in the final dataset. This sample size reflects all valid responses obtained during the recruitment period and constitutes a convenience sampling approach without a predetermined target size.

B. PROCEDURE

~~A. DATA COLLECTION~~

~~Data were collected through an online survey from July 1 to September 30, 2024. Participants completed the 60-item Japanese version of the HEXACO-PI-R [34]. Responses were only accepted if all items were answered, resulting in valid data from 13,668 participants. The survey was conducted in compliance with the conditions stipulated by hexaco.org, which include ensuring that the survey is not indexed by search engines and is accessible only via a password-protected form. The HEXACO-PI-R consists of six domains, each subdivided into four facets:~~

Participants accessed the survey via a publicly available URL shared on SNS platforms. Upon opening the survey, they were presented with an explanation of the study purpose, anonymity, voluntary participation, and data handling procedures. All survey items were set as mandatory in Google Forms, ensuring complete responses with no missing data. As participation was voluntary, participants could discontinue the survey at any time prior to submission. No incentives, rewards, or compensation were offered for participation.

Data collection was fully automated through the Google Forms system. As such, no direct interaction occurred between participants and researchers or interviewers at any stage of the data collection process.

C. MEASURES

Personality traits were assessed using the HEXACO-60 [39], a validated 60-item inventory measuring six core personality dimensions: Honesty-Humility~~: Sincerity, Fairness, Greed Avoidance, Modesty~~, Emotionality~~: Fearfulness, Anxiety, Dependence, Sentimentality~~, Extraversion~~: Social Self-Esteem, Social Boldness, Sociability, Liveliness~~, Agreeableness~~: Forgiveness, Gentleness, Flexibility, Patience~~, Conscientiousness~~: Organization, Diligence, Perfectionism, Prudence~~, and Openness to Experience~~: Aesthetic Appreciation, Inquisitiveness, Creativity, Unconventionality~~

~~Participants~~. Each item was rated ~~each item~~ on a 5five-point Likert scale ~~(1 = Strongly Disagree, 5 = Strongly Agree). Only anonymous response data were collected; demographic information such as gender and age was not obtained~~ranging from 1 ("strongly disagree") to 5 ("strongly agree"). Mean scores were computed for each factor and used in subsequent analyses.

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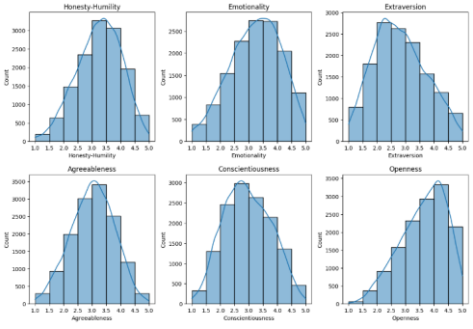
D. DATA SCREENING AND EXCLUSION CRITERIA

Several procedures were implemented to ensure data quality. Responses exhibiting signs of careless or inattentive responding—such as selecting the same option consecutively across a large number of items—were excluded. Responses with completion times that were extremely short or excessively long, relative to the distribution of survey completion times, were also excluded. Duplicate submissions were identified and removed using Google account authentication logs and metadata. Because all items were mandatory within the Google Forms system, the dataset contained no missing values or incomplete questionnaires.

E. ANALYTIC STRATEGY



All statistical analyses were conducted using Python 3.12.3. Descriptive statistics and correlation analyses were computed for all HEXACO dimensions. To identify personality prototypes, we applied multiple clustering approaches to enhance robustness and replicability.

First, hierarchical clustering using Ward's method was performed to explore the underlying structure and determine candidate numbers of clusters. This was followed by k-means clustering initialized from the Ward solution to refine cluster assignments. In addition, Latent Profile Analysis (LPA) and spectral clustering were conducted to evaluate the stability and interpretability of cluster solutions across alternative modeling frameworks. LPA models were estimated using Gaussian mixture modeling with diagonal covariance matrices, following standard practice in personality prototype research.

Cluster validity was assessed using internal fit indices (e.g., C-index, Dunn index, Silhouette coefficient, S_Dbw) and external validation procedures involving cross-sample classification with support vector machines (SVMs). All validity indices were computed on standardized trait scores to ensure comparability across clustering methods. The final seven-cluster solution was selected based on its consistency across subsamples, superior internal validity, and conceptual interpretability within the HEXACO framework.

F. ETHICAL CONSIDERATIONS

This study adhered to ethical guidelines specified in the Japanese Ministry of Health, Labour and Welfare (MHLW) regulations for research using anonymous, non-sensitive behavioral data (Chapter 1, Part 3, Section C). As the study did not involve identifiable personal information, clinical interventions, or sensitive categories, formal institutional review board (IRB) approval was not required. Participation was voluntary, anonymous, and based on informed consent obtained at the beginning of the survey. No identifying information (e.g., IP addresses, names, email addresses) was collected.

IV. RESULTS

~~B.~~A. DESCRIPTIVE STATISTICS

~~We calculated descriptive statistics, including median, mean, standard deviation, variance, skewness, and kurtosis for each domain and facet (Table 1). These statistics provide key insights into the distribution of personality traits among the sample. For example, the median and mean scores for Honesty-Humility and Emotionality were around 3.30, suggesting a relatively consistent pattern in these traits among Japanese participants. Standard deviations were .79 and .89, indicating moderate variability across participants, but without extreme deviations. In contrast, the median score for Extraversion was 2.70, with a mean of 2.80, suggesting lower levels of extraversion within the sample. The standard deviation of .91 indicated greater variability in extraversion compared to other traits. The median score for Openness to Experience was 3.70, with a mean of 3.62, which was notably higher than other domains, suggesting that Japanese participants exhibit a relatively open attitude toward new experiences and creative activities. The skewness and kurtosis values further revealed the shape of the trait distributions. For example, Extraversion and Emotionality exhibited flatter distributions with kurtosis values of -.64 and -.56, respectively. Meanwhile, Honesty-Humility showed a slightly left-skewed distribution with a skewness of -.26. The histograms for each domain provide a visual representation of the distribution of scores (Figure 1). Honesty-Humility and Emotionality were clustered around mid-range values, suggesting that participants generally exhibited balanced attitudes in these traits. In contrast, Extraversion scores were more concentrated in the lower range, indicating that many participants scored low on extraversion. This distribution highlights a common tendency toward introversion among the Japanese sample.~~

Detailed descriptive statistics for the six HEXACO personality dimensions are provided in Supplementary Tables S1–S2. All domains exhibited sufficient variability, with means close to the scale midpoint and standard deviations ranging between approximately 0.7 and 1.0. Skewness and kurtosis values indicated that several traits deviated from normality, consistent with previous research using HEXACO measures. Given these distributional characteristics, nonparametric analyses (see Section B) were deemed appropriate for the subsequent correlation tests. Moving these descriptive results to the supplementary file improves readability and allows the main text to focus on the cluster identification and validation analyses.

~~TABLE 1~~

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~~*DESCRIPTIVE STATISTICS OF ALL DOMAINS*~~

~~Domains~~	~~*Mdn*~~	~~*M*~~	~~*SD*~~	~~*Var*~~	~~*Skew*~~	~~*Kurt*~~
~~Honesty-Humility~~	~~3.30~~	~~3.24~~	~~.79~~	~~.62~~	~~-.26~~	~~-.37~~
~~Emotionality~~	~~3.30~~	~~3.23~~	~~.89~~	~~.79~~	~~-.25~~	~~-.56~~
~~Extraversion~~	~~2.70~~	~~2.80~~	~~.91~~	~~.83~~	~~.27~~	~~-.64~~
~~Agreeableness~~	~~3.00~~	~~3.01~~	~~.76~~	~~.57~~	~~-.11~~	~~-.36~~
~~Conscientiousness~~	~~2.90~~	~~2.95~~	~~.82~~	~~.68~~	~~.11~~	~~-.65~~
~~Openness~~	~~3.70~~	~~3.62~~	~~.81~~	~~.66~~	~~-.51~~	~~-.39~~

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~~*DESCRIPTIVE STATISTICS OF ALL FACETS*~~

~~Facets~~	~~*Mdn*~~	~~*M*~~	~~*SD*~~	~~*Var*~~	~~*Skew*~~	~~*Kurt*~~
~~Sincerity~~	~~3.00~~	~~2.92~~	~~1.11~~	~~1.24~~	~~.07~~	~~-.90~~
~~Fairness~~	~~3.67~~	~~3.53~~	~~1.29~~	~~1.67~~	~~-.48~~	~~-1.04~~
~~Greed-Avoidance~~	~~3.00~~	~~3.05~~	~~1.13~~	~~1.29~~	~~-.05~~	~~-.87~~
~~Modesty~~	~~3.50~~	~~3.43~~	~~1.16~~	~~1.34~~	~~-.44~~	~~-.74~~
~~Fearfulness~~	~~3.33~~	~~3.37~~	~~.98~~	~~.96~~	~~-.36~~	~~-.47~~
~~Anxiety~~	~~3.50~~	~~3.33~~	~~1.30~~	~~1.69~~	~~-.36~~	~~-1.11~~
~~Dependence~~	~~3.00~~	~~2.99~~	~~1.18~~	~~1.38~~	~~-.06~~	~~-.95~~
~~Sentimentality~~	~~3.33~~	~~3.14~~	~~1.11~~	~~1.24~~	~~-.11~~	~~-.94~~
~~Social Self-Esteem~~	~~3.00~~	~~2.95~~	~~1.14~~	~~1.30~~	~~.02~~	~~-.97~~
~~Social-Boldness~~	~~2.67~~	~~2.80~~	~~1.16~~	~~1.35~~	~~.17~~	~~-1.02~~
~~Sociability~~	~~2.00~~	~~2.47~~	~~1.24~~	~~1.54~~	~~.50~~	~~-.87~~
~~Liveliness~~	~~3.00~~	~~2.86~~	~~1.16~~	~~1.34~~	~~.11~~	~~-.94~~
~~Forgiveness~~	~~2.00~~	~~2.09~~	~~1.08~~	~~1.16~~	~~.84~~	~~-.15~~

Gentleness	3.33	3.16	1.05	1.11	-.20	-.78
Flexibility	3.00	3.08	.93	.87	-.11	-.54
Patience	3.50	3.56	1.16	1.33	-.48	-.76
Organization	2.67	2.74	1.17	1.37	.24	-.92
Diligence	3.00	3.11	1.08	1.16	-.12	-.76
Perfectionism	3.33	3.19	1.02	1.03	-.19	-.76
Prudence	2.67	2.71	1.05	1.10	.24	-.80
Aesthetic-Appreciation	4.00	3.59	1.21	1.46	-.64	-.64
Inquisitiveness	3.50	3.29	1.23	1.51	-.34	-.92
Creativity	3.67	3.40	1.16	1.35	-.41	-.87
Unconventionality	4.33	4.04	.84	.71	-.81	.13

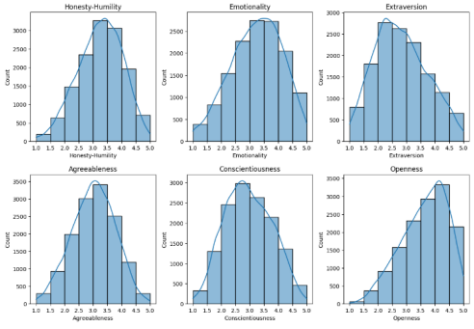

Figure 1

FIGURE 1. Histograms showing the distributions of scores for the six HEXACO domains among 13,668 Japanese participants. Honesty–Humility and Emotionality were centered around mid-range values, while Extraversion was skewed toward lower scores, indicating a general tendency toward introversion.

B. ~~CORRELATION AND~~ RELIABILITY ~~OF~~ESTIMATES

~~C.~~ Internal consistency coefficients (Cronbach's α) were calculated for each of the six HEXACO ~~TRAITS~~dimensions. All domains demonstrated acceptable to good reliability, with α coefficients ranging from .78 to .84, which aligns with previously reported values for the HEXACO-60 instrument. These results indicate that the measurement model was sufficiently reliable for use in the subsequent correlational and clustering analyses.

C. NORMALITY TESTS

Shapiro–Wilk tests were conducted to examine the distributional properties of the six HEXACO domains. All traits showed statistically significant deviations from normality:

- Honesty–Humility: W = 0.9907, p < .001
- Emotionality: W = 0.9863, p < .001
- Extraversion: W = 0.9826, p < .001
- Agreeableness: W = 0.9952, p < .001
- Conscientiousness: W = 0.9897, p < .001
- Openness: W = 0.9697, p < .001

Given these results and the large sample size, which increases sensitivity to even minor deviations from normality, subsequent correlational analyses were performed using Spearman's rank-order correlation, a nonparametric alternative robust to non-normal distributions.

D. CORRELATION ANALYSIS

Spearman's rank-order correlations were computed among the six HEXACO domains, with 95% confidence intervals included in Table 1. Overall, the observed associations were consistent with theoretical expectations of the HEXACO framework.

Honesty–Humility showed small positive correlations with Agreeableness and Conscientiousness, and a small negative correlation with Extraversion. Emotionality demonstrated a small negative association with Extraversion, along with small positive associations with Agreeableness and Conscientiousness. Extraversion exhibited small positive correlations with Openness and Agreeableness. Agreeableness and Conscientiousness were largely independent, showing only a negligible negative association. Finally, Openness showed small positive correlations with Extraversion, Agreeableness, and Conscientiousness.

All reported p-values were below the significance thresholds used in this study (see Table 1 notes for thresholds). Full correlation coefficients and their 95% confidence intervals are presented in Table 1.

TABLE 1

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~~The assumption of normality was examined using the Shapiro–Wilk test. As most variables significantly deviated from a normal distribution, nonparametric Spearman's rank correlations were conducted to examine the associations among HEXACO traits (Table 2). Several notable relationships emerged. First, there was a negative correlation between Honesty–Humility and Extraversion (r = -.2313), indicating that individuals with higher levels of honesty and humility tend to score lower on extraversion, possibly reflecting the influence of Japanese cultural norms that emphasize modesty and restraint. Conversely, Honesty–Humility was positively correlated with Conscientiousness (r =.1835), suggesting that individuals who are honest and humble are also likely to be more organized and disciplined. This relationship may reflect the close connection between integrity and responsibility in Japanese culture. In terms of reliability, Cronbach's alpha was calculated for the HEXACO scales. The overall scale had an alpha of .64, indicating moderate internal consistency. Domain-specific alphas were: Honesty–Humility (.57), Emotionality (.79), Extraversion (.79), Agreeableness (.68), Conscientiousness (.76), and Openness (.72). While Emotionality and Extraversion demonstrated strong internal consistency, Honesty–Humility~~

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had a relatively low alpha of .57, suggesting that this domain may require further refinement for enhanced reliability in future research.

TABLE 2

INTER-TRAIT CORRELATIONS (HEXACO DOMAINS)

HEXACO Domains	Honesty-Humility HH	Emotionality EM	Extraversion EX	Agreeableness AG	Conscientiousness CO	Openness OE
Honesty-Humility HH	1.00					
Emotionality EM	.04** [.024, .058]	1.00				
Extraversion EX	-.24** [-.247, -.214]	-.21** [-.217, -.184]	1.00			
Agreeableness AG	.19** [.162, .194]	.10** [.088, .122]	.01**	.04** [.024, .058]	1.00	
Conscientiousness CO	.18** [.167, .021]	.11** [.092, .125]	-.05** [-.062, -.027]	-.04**	-.06** [-.071, -.037]	-.05*	1.00
Openness OE	.06** [.041, .075]	.04** [.007, .042]	.02** [.069, .103]	.08** [.055, .089]	.09**	.08**	.07**	.18** [.159, .191]	1.00

Note. Values represent Spearman correlations with 95% confidence intervals shown in brackets. HH = Honesty–Humility; EM = Emotionality; EX = Extraversion; AG = Agreeableness; CO = Conscientiousness; OE = Openness to Experience.

**p < .01; *p < .05.

D.E. DERIVATIONDETERMINATION OF THE PROTOTYPESCLUSTER NUMBER

Determining the optimal number of clusters is a critical step in prototype derivation. In the present study, multiple internal and external validation criteria were employed to ensure the robustness and replicability of the cluster solution. Internal validity was evaluated using commonly adopted indices such as the silhouette coefficient, Dunn index, S_Dbw index, and the C index, all computed on standardized trait scores. These indices consistently indicated that solutions between six and eight clusters provided superior separation and cohesion compared with more parsimonious or more granular structures.

To complement these internal fit measures, external validation procedures were conducted using cross-sample classification with support vector machines (SVMs). This approach assesses the extent to which cluster assignments generalize beyond the original estimation sample, capturing non-spherical or non-linear boundaries that traditional methods may overlook. Solutions with seven clusters demonstrated the highest classification accuracy, supporting their stability.

In addition to these procedures, we incorporated a comprehensive set of internal and external indices described in prior clustering literature. For internal evaluation, we applied (1) elbow-curve inspection, (2) cost-based information criteria including AIC and BIC [41], (3) density-based indices such as S_Dbw [42], and (4) geometric or structural indices including the C index [43], Generalized Dunn index [44], Baker–Hubert Gamma index [45], G⁺ index [46], and the Silhouette index [47]. For external validity, agreement among clustering methods was assessed using Cohen's kappa and the Hubert–Arabie Adjusted Rand Index [40], demonstrating strong concordance particularly for the seven-cluster solution.

Taken together, these internal and external indices consistently supported the seven-cluster structure as the most stable, interpretable, and replicable solution.

TABLE 2

In this study, we followed the approach of Kerber et al. [30], employing multiple validity criteria and clustering algorithms. Specifically, we used three different clustering techniques: a) Ward's method followed by k-means clustering (hierarchical clustering), b) latent profile analysis (LPA), and c) spectral clustering. The Ward's method and k-means clustering combination is a widely used approach in personality type research. It begins with hierarchical clustering via Ward's method, which is then used as the initial values for k-means clustering. This method reduces the risk of local optima and allows for efficient clustering. Latent profile analysis (LPA) is a probabilistic finite mixture modeling technique. It assumes that latent classes or profiles exist behind the observed variables and calculates the probability that each individual belongs to a given class. This method allows for more flexible modeling compared to traditional methods and is particularly suitable for continuous variables. Spectral clustering uses graph theory to detect non-linearly separable clusters. It constructs a similarity graph among data points and uses the eigenvectors of the graph's Laplacian matrix to perform clustering. This method is useful for identifying clusters with complex shapes that might not be captured by linear approaches. By employing these three methods, we compared clustering results across different approaches to identify robust personality types.

E. DETERMINING THE NUMBER OF CLUSTERS

Determining the optimal number of clusters is a crucial step in clustering analysis. In this study, we used both internal and external validity indices to decide the optimal number of clusters. For external validity, we used Cohen's kappa and the Hubert–Arabie adjusted Rand index [35] to evaluate the agreement between different clustering results. A support vector machine (SVM) classifier was also used to assign clusters, allowing for more flexible cluster assignment that accommodates non-spherical distributions. For internal validity, we employed the following indices: (1) the elbow method, (2) cost function-based criteria (AIC and BIC [36]), (3) cluster density-based criteria (S_Dbw index [37]), and (4)

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geometric assumption based indices, including the C index [38], Generalized Dunn index [39], Baker Hubert Gamma index [40], G+ index [41], and Silhouette index [42]. These diverse indices provided a comprehensive evaluation of the validity of the cluster solutions.

F.A. PROCEDURE

Following Kerber et al. [30], the following procedure was used to identify and evaluate personality types:

1. Ten random subsamples were created from the full dataset (N = 13,668), and each subsample was split into two halves.
2. Three clustering algorithms (Ward's method + k-means, LPA, and spectral clustering) were applied to each half, yielding 3-cluster, 5-cluster, and 7-cluster solutions.
3. SVM classifiers were used to reassign the other half of the subsample to clusters.
4. Cohen's kappa and the adjusted Rand index (ARI) were calculated to evaluate the consistency of the clustering results.
5. The results were averaged across the 10 subsamples to obtain an overall evaluation for each algorithm and cluster number.
6. Internal validity indices (described in section 2.7) were calculated to assess the internal consistency of the cluster solution.
7. The results of the external and internal validity indices were combined to determine the optimal number of clusters and clustering algorithm.
8. The full dataset was clustered using the selected optimal method and number of clusters.
9. Labeling of each cluster was conducted based on the characteristics of the profiles in reference to previous studies.

This procedure enabled us to identify and evaluate HEXACO personality types in the Japanese sample. The next section provides a detailed report of the results.

IV.I. RESULTS

A. INTERNAL AND EXTERNAL CLUSTER FIT INDICES

In evaluating the clustering solutions, both internal and external validity indices were used to assess the fit of various clustering methods. Results from the elbow method (Table 3) showed that the cost function decreased gradually as the number of clusters increased across all methods. However, the reduction became less pronounced at a certain point, which is typically used to determine the optimal number of clusters. Specifically, for Ward's method followed by k-means clustering, there was a significant reduction in the cost function up to seven clusters (47505.87 → 43827.07), after which the reduction slowed. Similarly, both the Gaussian Mixture Model (LPA) and spectral clustering also showed a clear turning point at seven clusters. For external validity, Cohen's kappa and the Hubert-Arabie adjusted Rand index (ARI) were used to evaluate the consistency of clustering results (Table 4). Internal validity indices such as AIC, BIC, S_Dbw, C-index, Generalized Dunn index, and Silhouette index were also calculated (Table 4).

TABLE 3

RESULTS OF ELBOW METHOD

Number of Clusters	Ward + KMeans	LPA (GaussianMixture)	SpectralClustering
3	61597.6	227241.4	62310.0
5	52890.6	227338.7	53836.0
7	47505.9	226641.4	49163.8
9	43827.1	226949.4	46377.4
11	41048.3	227150.8	43682.6
13	38998.6	227291.1	41077.8

TABLE 43

RESULTS OF INTERNAL AND EXTERNAL CLUSTER FIT INDICES

validity criterion	3 cluster			5 cluster			7 cluster			Interpretation
	ward_kmeans	gmm	spectral	ward_kmeans	gmm	spectral	ward_kmeans	gmm	spectral	
C-index	.28	.29	.29	.24	.27	.25	.21	.25	.23	min
Generalized Dunn Index	.015	.019	.015	.021	.018	.022	.022	.018	.000	max
Baker-Hubert Gamma	.41	.37	.37	.49	.40	.46	.55	.47	.52	max
G(+)	.49	.44	.47	.63	.56	.60	.70	.63	.67	max
Silhouette	.144	.117	.132	.129	.090	.119	.127	.080	.111	max
S_Dbw	1.68	1.61	1.72	1.74	1.67	1.74	1.75	1.68	1.71	min
AIC	61625.8	226617.0	62346.0	52841.5	226293.0	53897.2	47507.5	225174.5	49247.8	min
BIC	61761.2	227241.4	62481.4	53067.2	227338.7	54122.9	47823.4	226641.4	49563.7	min
Cohen's kappa	.12	.10	.01	.42	.35	.03	.94	.91	.00	max
Adjusted Rand Index (ARI)	.24	.25	.18	.41	.36	.28	.88	.83	.38	max

Internal Validity Indices (Table 4): The C-index showed a decrease in value as the number of clusters increased across all methods. Particularly for the 7-cluster solution, Ward's method followed by k-means showed a value of .21, Gaussian Mixture Model (GMM) showed .25, and spectral clustering showed .23, indicating that the 7-cluster solution had better clustering performance than other solutions. The Generalized Dunn Index showed lower values for the 3-cluster solution (.015–.019), but slightly higher values for the 5-cluster and 7-

~~cluster solutions (5-cluster: .018–.022, 7-cluster: .018–.022). Notably, the Ward's method followed by k-means showed the best performance for the 7-cluster solution. Results from the Baker-Hubert Gamma index and the G(+) index also indicated that the 7-cluster solution using Ward's method followed by k-means performed the best. For the Baker-Hubert Gamma index, Ward's method followed by k-means yielded a value of .55, GMM showed .47, and spectral clustering showed .52. For the G(+) index, Ward's method followed by k-means achieved a high score of .70. The Silhouette index showed relatively lower results overall, with Ward's method followed by k-means showing the highest value of .144 for the 3-cluster solution. Other methods showed values ranging from .080 to .111, with Ward's method + k-means exhibiting relatively better clustering performance. The S_Dbw index values ranged from 1.61 to 1.75 across all methods, with a slight increase as the number of clusters increased. The GMM 3-cluster solution showed the lowest value of 1.61.~~

~~**External Validity Indices (Table 4):** For Cohen's kappa, the 7-cluster solution using Ward's method followed by k-means showed the highest value (.9388), with GMM also yielding a high consistency score (.9050). In contrast, spectral clustering produced a low score of -.0011 for the 7-cluster solution, indicating that Ward's method followed by k-means and GMM were the most reliable clustering methods for this solution. The Adjusted Rand Index (ARI) also showed strong results for the 7-cluster solution using Ward's method followed by k-means (.8829) and GMM (.8264). In contrast, spectral clustering yielded a lower ARI of .3817, further supporting the superior performance of Ward's method followed by k-means and GMM.~~

~~In conclusion, the results of both external and internal validity indices indicated that the 7-cluster solution using Ward's method followed by k-means provided the best overall performance. Cohen's kappa and ARI scores were particularly high, confirming the consistency and stability of this clustering method. GMM also performed well in the 7-cluster solution, suggesting it as another viable clustering method.~~

~~*B. CLUSTERING AND LABELING*~~

~~Using the 7-cluster solution from the k-means method, the average scores of each cluster on the HEXACO traits were calculated (Figure 2). Additionally, the percentage of participants in each cluster was determined, and the clusters were labeled based on their profiles.~~

~~- Cluster 1: Honesty-Humility (3.3), Emotionality (2.7), Extraversion (2.44), Agreeableness (2.87), Conscientiousness (2.41), Openness (2.48). This cluster is characterized by introversion, agreeableness, and slightly lower self-management skills. Comprising 13.81% of the sample, it was labeled as the "Reserved Type." Individuals in this group are modest and prioritize social harmony but may face challenges in self-management.~~

~~- Cluster 2: Honesty-Humility (3.0), Emotionality (3.93), Extraversion (2.16), Agreeableness (2.98), Conscientiousness (2.41), Openness (3.62). This group exhibits high emotional sensitivity and low extraversion but is highly open-minded. Representing 16.96% of the sample, it was labeled as the "Emotionally Sensitive Type." Individuals here express emotions openly and possess a strong sense of creativity and openness to new experiences.~~

~~- Cluster 3: Honesty-Humility (3.82), Emotionality (3.9), Extraversion (2.06), Agreeableness (3.15), Conscientiousness (3.72), Openness (3.84). This cluster consists of highly conscientious individuals who are also introverted and exhibit strong intellectual curiosity. With 17.3% of the sample, it was labeled as the "Conscientious Introvert Type." These individuals demonstrate high levels of honesty and responsibility, with a preference for solitary and reflective activities.~~

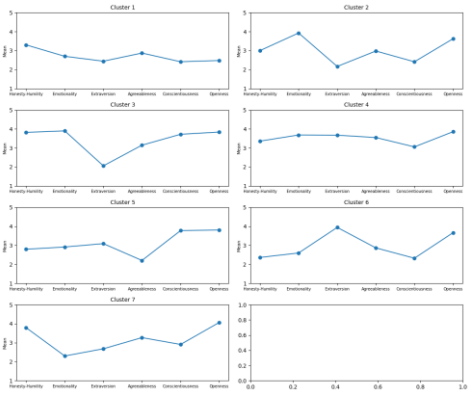
~~- Cluster 4: Honesty-Humility (3.36), Emotionality (3.68), Extraversion (3.67), Agreeableness (3.55), Conscientiousness (3.06), Openness (3.85). Individuals in this cluster are sociable, agreeable, and excel in social interactions. Representing 15.47% of the sample, this group was labeled the "Extraverted Agreeable Type." These individuals are outgoing, cooperative, and socially engaged.~~

~~- Cluster 5: Honesty-Humility (2.79), Emotionality (2.91), Extraversion (3.09), Agreeableness (2.2), Conscientiousness (3.78), Openness (3.81). This group is characterized by high self-management but lower agreeableness. Representing 11.85% of the sample, this cluster was labeled as the "Independent Type." These individuals exhibit strong self-discipline and independence but may struggle with maintaining harmonious relationships with others.~~

~~- Cluster 6: Honesty-Humility (2.36), Emotionality (2.6), Extraversion (3.94), Agreeableness (2.86), Conscientiousness (2.32), Openness (3.66). This group is sociable and emotionally stable but exhibits lower self-management skills. With 11.64% of the sample, this cluster was labeled as the "Extraverted Stable Type." These individuals are confident and emotionally composed, with high social energy.~~

~~- Cluster 7: Honesty-Humility (3.79), Emotionality (2.3), Extraversion (2.68), Agreeableness (3.27), Conscientiousness (2.91), Openness (4.06). This group is introverted but highly conscientious and open to new experiences, often demonstrating leadership skills. Representing 12.97% of the sample, this group was labeled as the "Introverted Leader Type." These individuals are honest, flexible thinkers, and possess leadership potential despite their introverted nature.~~

~~Each of these clusters reflects distinct personality profiles based on the HEXACO model and highlights the diversity of personality types within the Japanese sample. These findings offer a nuanced understanding of individual personality tendencies within the context of Japanese culture.~~



~~Figure 2~~

~~F. Mean HEXACO trait profiles for the seven~~CLUSTER PROTOTYPES

The seven-cluster solution produced distinct and interpretable personality prototypes, each reflecting unique configurations of HEXACO traits. Mean scores for all traits across clusters are presented in Table 3, and visualized in Figure 2.

Cluster 1 "Reserved" was characterized by above-average Honesty–Humility and Agreeableness, combined with moderate levels of Emotionality and Conscientiousness. This profile corresponds to a generally prosocial and cooperative pattern.

Cluster 2 " Emotionally Sensitive" showed elevated Emotionality and lower Extraversion, reflecting a more introverted and affectively sensitive disposition. Agreeableness and Honesty–Humility were moderate, suggesting a quiet but interpersonally considerate profile.

Cluster 3 "Exploratory Extravert" exhibited above-average Extraversion and Openness, with moderate Agreeableness and Conscientiousness. This constellation represents an outgoing, explorative, and cognitively flexible personality style.

Cluster 4 " Conscientious Introvert" displayed high Conscientiousness and moderate Openness, accompanied by average levels of the remaining traits. This cluster resembles a diligent and self-regulated prototype.

Cluster 5 "Self-Oriented Independent" demonstrated lower scores on Agreeableness and Honesty–Humility relative to other clusters, alongside moderate Extraversion and Openness. This pattern may indicate a more competitive and assertive interpersonal style.

Cluster 6 " Emotionally Volatile Extravert" was characterized by elevated Emotionality combined with relatively lower Agreeableness, indicating a more affectively reactive and potentially conflict-prone interpersonal style.

Cluster 7 " Reliable Introvert" combined high Agreeableness with moderate-to-high Conscientiousness and Emotionality, reflecting an interpersonally warm and dependable profile.

Together, these seven prototypes capture a broad range of personality configurations present in the Japanese population. Their distinctiveness and interpretability support the validity of the clustering solution and align with prior research demonstrating that heterogeneous personality ~~clusters identified~~ structures can be parsimoniously represented through ~~k-means analysis. Each line represents unstandardized means across the six domains. The profiles reveal distinct~~ discrete profile patterns.

Figure 2

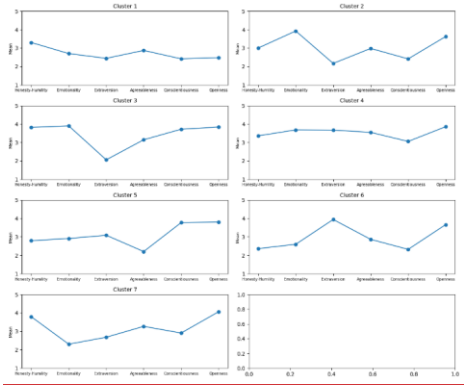


FIGURE 2. Radar plot illustrating the mean HEXACO domain scores for the seven identified personality ~~patterns such as~~prototypes: Reserved, Emotionally Sensitive, Exploratory Extravert, Conscientious Introvert, ~~Extraverted Agreeable,~~Self-Oriented Independent, ~~Extraverted Stable, and Introverted Leader,~~Emotionally Volatile Extravert, and Reliable Introvert. Each colored line represents standardized scores across the six HEXACO dimensions (Honesty–Humility, Emotionality, Extraversion, Agreeableness, Conscientiousness, Openness to Experience), highlighting ~~cultural variation in Japanese personality structure~~distinct configurational patterns that differentiate the prototypes.

V. DISCUSSION

A. ~~SUMMARY~~OVERVIEW OF ~~RESULTS~~KEY FINDINGS

~~This~~The present study ~~clustered over 13,000 personality data points from Japanese individuals based on the HEXACO model, identifying~~ identified seven distinct personality ~~types. The combination of Ward's~~ prototypes in a large Japanese sample using the HEXACO framework and a multi-method ~~and k-means~~ clustering ~~showed~~strategy. By integrating hierarchical clustering with k-means, latent profile analysis, and spectral clustering, the ~~best fit in terms of~~ results demonstrated a stable and replicable seven-cluster solution that was consistently supported by both internal and external validity~~, with~~ indices. This finding extends prior personality prototype research, which has typically reported four to five profiles in Western samples, by showing that a more differentiated structure emerges when culturally

specific trait covariation is examined in a non-WEIRD context [27], [28], [35].

Across the identified prototypes, several patterns aligned with previously reported personality types, such as profiles resembling resilient or reserved configurations characterized by relatively high Agreeableness and Conscientiousness. At the same time, other prototypes reflected combinations of traits that appear particularly ~~high scores for Cohen's kappa and the adjusted Rand index. The seven identified clusters are as follows:~~salient within the Japanese cultural context, including profiles marked by emotional sensitivity combined with introversion, as well as dependable but interpersonally restrained patterns. These results support the notion that personality structure reflects both universal psychological processes and culturally shaped norms regarding emotional expression, social harmony, and self-regulation [36], [37]. Importantly, the presence of multiple introversion-oriented yet functionally adaptive profiles suggests that low Extraversion does not uniformly indicate maladaptive functioning in East Asian cultural settings.

~~1. Reserved Type: Introverted, agreeable, but with lower self-management skills.~~

~~2. Emotionally Sensitive Type: Emotionally sensitive, low extraversion, but highly open-minded.~~

~~3. Conscientious Introvert Type: Honest, conscientious, introverted individuals with high responsibility.~~

~~4. Extraverted Agreeable Type: Outgoing and highly cooperative in social settings.~~

~~5. Independent Type: Self-disciplined and independent, with lower agreeableness.~~

~~6. Extraverted Stable Type: Sociable and emotionally stable but lower in self-management.~~

~~7. Introverted Leader Type: Honest, introverted, and open to new experiences, with leadership potential.~~

~~These clusters reflect various combinations of HEXACO traits, highlighting the diversity of personality types within Japanese individuals. Some of the clusters align with previously reported personality types, while others reveal characteristics unique to Japanese culture.~~

A central contribution of this study lies in demonstrating the prominent role of the Honesty–Humility dimension in differentiating personality prototypes within the Japanese population. Variations in Honesty–Humility meaningfully distinguished cooperative, reliable profiles from more self-oriented or emotionally volatile patterns, highlighting its importance beyond trait-level analyses. This finding is consistent with prior work emphasizing the cultural and moral relevance of Honesty–Humility in shaping social behavior and interpersonal functioning [18], [21]. Taken together, the results indicate that HEXACO-based personality prototypes capture both cross-culturally recurrent patterns and culturally specific configurations, thereby advancing the understanding of personality structure in non-Western contexts and addressing a key gap identified in previous research [25], [35].

B. CULTURAL INTERPRETATIONS OF ~~FINDINGS~~THE PROTOTYPES

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~~The results of this study provide novel insights into personality types in Japan, aligning with some aspects of previous research while also identifying personality types that are specific to Japanese culture. For example, the Reserved Type and Conscientious Introvert Type correspond to overcontrolled and resilient types reported in previous studies. However, these types also reflect a strong emphasis on introversion and social harmony, which are more pronounced in Japanese culture. The Emotionally Sensitive Type and Introverted Leader Type also stand out, reflecting a greater sensitivity to emotional nuances and a more reserved form of leadership that is valued in Japan. These findings support the research questions proposed in section 1.5, particularly the first question regarding whether Japanese personality types align with those identified in previous studies. While the results confirm the existence of universal personality structures, the seven clusters identified in this study highlight unique cultural traits that differentiate Japanese individuals from those in Western studies.~~

~~*C. CULTURAL INFLUENCES ON PERSONALITY*~~

~~The findings of this study reaffirm the importance of cultural factors in shaping personality types, as highlighted in previous personality research. While some personality types, such as the Resilient and Overcontrolled types, were replicated in the Japanese sample, other clusters demonstrated unique cultural influences on personality formation. For instance, the Reserved Type and Conscientious Introvert Type reflect Japanese cultural norms that value introversion and modesty, traits that are culturally reinforced through societal expectations of politeness and social harmony. These traits stand in contrast to more extroverted personality profiles that are more prevalent in Western societies. On the other hand, clusters such as the Extraverted Stable Type and the Independent Type, characterized by lower levels of Agreeableness or higher levels of Extraversion, reflect a shift towards individualism in modern Japanese society, diverging from traditional collectivist norms. Additionally, the high prevalence of Emotional Sensitivity and Introverted Leadership in certain clusters highlights the nuanced role that emotional expressiveness and leadership play in Japanese culture. These traits suggest that, while emotional regulation and humility are often emphasized, there is room for creative and emotionally sensitive leadership within the Japanese social context. In summary, this study illustrates how cultural values, such as humility, social harmony, and emotional sensitivity, are embedded in Japanese personality structures, influencing the development and manifestation of personality types.~~

~~*D. LIMITATIONS OF THE STUDY*~~

~~While this study provides important insights into the personality structure of Japanese individuals using the HEXACO model, several limitations must be acknowledged.~~

~~These limitations should be considered when interpreting the results and in the design of future research. First, the lack of demographic data in this study, such as gender, age, and geographic region, restricts the ability to examine how these factors may influence the formation and characteristics of personality types. Including such demographic information could provide a more granular understanding of personality types and their development across different population subgroups. Future studies could address this limitation by analyzing how personality types vary across demographic categories within the Japanese population. Second, this study did not examine the relationships between personality types and other important psychological or behavioral variables, such as physical and mental health, well-being, self-esteem, or risk-taking behavior. These variables are known to be closely linked to personality traits and could offer a deeper understanding of how personality types affect various life outcomes. Future research could incorporate these factors to develop a more comprehensive picture of personality types and their real-world implications. Third, the internal consistency of the Honesty–Humility domain, as indicated by its relatively low Cronbach's alpha (.57), suggests that this domain may require further refinement. This raises the question of whether the measurement of this domain accurately captures the nuances of Honesty–Humility in Japanese culture. Future studies could improve the scale's reliability by refining the items or adapting them to better fit the cultural context. Additionally, the cross-cultural comparison of HEXACO personality types remains a future research challenge. While this study focused solely on Japanese individuals, comparing these findings with data from other non-Western and Western cultures would provide more insight into the universality and cultural specificity of personality types. Such comparative studies could help clarify how cultural norms and values influence personality structures across different societies. Finally, the study's use of an internet-based survey may have introduced sampling biases, potentially over-representing individuals who are more comfortable using digital platforms. This could affect the generalizability of the findings to the broader Japanese population. Future studies should consider more diverse data collection methods to obtain a representative sample. Despite these limitations, this study contributes significantly to understanding Japanese personality structures based on the HEXACO model and highlights the cultural factors that influence personality formation.~~

~~*E. IMPLICATIONS FOR FUTURE RESEARCH*~~

~~The findings of this study offer several directions for future research. First, future studies should aim to address the limitations mentioned above, particularly by incorporating demographic variables and behavioral indicators such as health and well-being. This would help clarify the role of personality types in different life domains and provide more practical applications for the findings. Moreover, cross-cultural studies using the HEXACO model would enable researchers to explore how cultural contexts shape personality types. This could lead to a more comprehensive understanding of the interplay between universal personality traits and cultural values, thus enriching the field of personality research. Lastly, the refinement of the Honesty–Humility domain within the Japanese context is an important future research direction. As this study indicated some reliability issues with this domain, further investigations into how it can be more accurately measured and understood in non-Western contexts will be essential for advancing the HEXACO model's applicability.~~

The seven personality prototypes identified in this study can be meaningfully interpreted through the lens of Japanese cultural norms emphasizing social harmony, emotional restraint, and context-sensitive self-regulation. Several profiles were characterized by relatively low Extraversion combined with moderate to high Agreeableness and Conscientiousness, reflecting interpersonal styles that prioritize cooperation, reliability, and indirect social engagement. In contrast to Western interpretations that often associate low Extraversion with social withdrawal, such configurations may represent culturally adaptive patterns in Japan, where reserved behavior and attentiveness to social context are socially valued and functionally effective [36].

Notably, the coexistence of emotionally sensitive and emotionally volatile profiles highlights the nuanced role of Emotionality within Japanese personality structure. While Emotionality has often been interpreted as a vulnerability-related trait in Western samples, its integration with Agreeableness or Conscientiousness in several prototypes suggests a culturally regulated form of emotional awareness rather than maladaptive reactivity. This interpretation aligns with cross-national HEXACO research demonstrating that mean levels and functional correlates of Emotionality differ substantially across cultural contexts, particularly in East Asian societies [48]. Thus, emotional sensitivity in Japan may serve affiliative and relational functions rather than reflecting dysregulation per se.

The Honesty–Humility dimension further underscores the cultural specificity of the identified prototypes. Profiles high in Honesty–Humility tended to co-occur with interpersonal reliability and social responsibility, reinforcing cultural ideals related to sincerity, modesty, and trustworthiness. At the same time, the presence of self-oriented profiles characterized by relatively lower Honesty–Humility suggests increasing heterogeneity within contemporary Japanese society, potentially reflecting shifts toward individualism in competitive social and occupational environments. Together, these findings indicate that Japanese personality prototypes are best understood not as deviations from Western norms, but as culturally grounded configurations shaped by enduring values alongside ongoing social change.

C. THEORETICAL IMPLICATIONS

The present findings offer several theoretical contributions to personality psychology, particularly in relation to cross-cultural personality structure and person-centered modeling.

First, this study demonstrates that HEXACO-based personality prototypes can exhibit greater differentiation in a non-Western context than has typically been reported in Western samples. Whereas previous prototype studies often converged on four to five profiles [27], [28], the present analysis consistently supported a seven-cluster solution across multiple validation methods. This suggests that culturally specific patterns of trait covariation may remain obscured when prototype models are derived primarily from Western populations, underscoring the importance of extending person-centered approaches to diverse cultural contexts.

Second, the results advance theoretical understanding of how universal personality dimensions interact with culture-specific norms to shape higher-order personality configurations. While several prototypes showed structural similarities to established types such as resilient or reserved profiles, their trait compositions differed in systematic ways—particularly with respect to Extraversion and Emotionality. These findings support a configurational view of personality in which the same trait levels may serve different functional roles depending on cultural context, rather than implying equivalent psychological meaning across societies [36]. From a theoretical standpoint, this challenges assumptions of direct generalizability from Western-derived prototype structures and highlights the need for culturally sensitive models of personality organization.

Third, the study clarifies the theoretical significance of the Honesty–Humility dimension in typological personality research. Beyond its established role in predicting socially relevant behaviors at the trait level [18], Honesty–Humility emerged as a key organizing dimension differentiating cooperative, reliable prototypes from more self-oriented configurations. This finding suggests that Honesty–Humility may function as a central axis in the formation of personality types, particularly in cultures where moral self-regulation and interpersonal trust are socially emphasized. Integrating Honesty–Humility into person-centered frameworks therefore represents a theoretical advancement over Big Five–based typologies, offering a more comprehensive account of how moral and social traits contribute to personality structure in culturally embedded contexts.

D. PRACTICAL / CLINICAL / IMPLICATIONS

The identification of culturally grounded personality prototypes has important implications for mental health assessment and psychological support in Japan. Person-centered profiles provide clinically intuitive frameworks that may complement traditional dimensional assessments by capturing how multiple traits co-occur within individuals. For example, prototypes characterized by elevated Emotionality combined with low Extraversion may reflect heightened sensitivity to stress and interpersonal demands, even in the absence of overt psychopathology. In clinical and counseling settings, such profiles could help practitioners identify individuals who may be at increased risk for anxiety-related difficulties or social withdrawal, particularly when exposed to sustained environmental stressors [12], [17].

Importantly, the present findings suggest that low Extraversion or high Emotionality should not be uniformly interpreted as maladaptive within the Japanese context. Several prototypes combined these traits with high Agreeableness or Conscientiousness, indicating forms of emotional sensitivity that may support empathy, responsibility, and relational attunement. From a clinical perspective, this highlights the risk of over-pathologizing culturally normative personality configurations when using assessment frameworks derived primarily from Western norms. Prototype-based interpretations may therefore aid clinicians in distinguishing culturally adaptive emotional styles from maladaptive patterns requiring intervention, thereby improving diagnostic accuracy and cultural sensitivity in mental health practice.

Furthermore, the differentiation of prototypes along the Honesty–Humility dimension has potential relevance for psychosocial adjustment and therapeutic engagement. Individuals characterized by higher Honesty–Humility and Agreeableness may show stronger therapeutic alliance, compliance with treatment recommendations, and prosocial coping strategies, whereas more self-oriented or emotionally volatile profiles may require tailored intervention approaches emphasizing emotion regulation, impulse control, and interpersonal skills training. In this sense, HEXACO-based personality prototypes could inform personalized prevention and intervention strategies, supporting a move toward culturally informed, person-centered mental health care that aligns with contemporary models of psychological adaptation and well-being.

E. LIMITATIONS

Several limitations of the present study should be acknowledged when interpreting the findings. First, although the sample size was large and provided substantial statistical power, the study employed a cross-sectional online survey design based on convenience sampling. Participants were recruited through social networking platforms, which may have resulted in sampling biases related to internet use, self-selection, or interest in personality-related content.

In addition, the seven-cluster solution identified in this study may appear more granular than the - to five-type structures commonly reported in Western personality prototype research. However, this differentiation is unlikely to reflect overfitting, as the seven-cluster structure was consistently supported by multiple clustering algorithms, internal validity indices, and external cross-sample classification procedures. Instead, the higher number of prototypes may reflect culturally specific patterns of trait covariation in the Japanese context, which are less apparent in Western samples.

Second, the study relied exclusively on self-report measures of personality traits. While the HEXACO-60 has demonstrated robust psychometric properties across cultures [18], self-report data are inherently subject to response biases

such as social desirability, self-presentation, and limited introspective accuracy. These issues may be especially relevant in cultural contexts that emphasize modesty and social harmony, potentially influencing the expression of traits such as Honesty–Humility or Emotionality. As a result, the observed personality prototypes should be interpreted as subjective self-representations rather than objective behavioral profiles.

Third, the present analysis focused solely on personality structure and did not include external criteria such as clinical symptoms, well-being indicators, or functional outcomes. This limits the ability to draw direct conclusions about the mental health relevance or adaptive value of each prototype. Although theoretical and prior empirical work suggests meaningful links between HEXACO traits and psychological adjustment [12], [17], the absence of concurrent mental health measures prevents definitive clinical validation of the identified profiles. Therefore, the clinical interpretations offered in this study should be regarded as provisional and hypothesis-generating rather than diagnostic conclusions.

F. FUTURE WORK

Future research should seek to replicate the present findings using more diverse sampling strategies and research designs. Probability-based sampling or stratified recruitment methods would help assess the generalizability of the seven-prototype structure beyond online convenience samples. In addition, replication across independent Japanese datasets would further strengthen confidence in the stability of the identified profiles. Given ongoing debates regarding the robustness of personality types, such replication efforts are essential for establishing a cumulative and reproducible body of evidence in person-centered personality research.

A second important direction involves cross-cultural comparison. Applying the same multi-method clustering framework to samples from other East Asian societies, as well as Western populations, would allow for direct tests of universality and cultural specificity in HEXACO-based personality prototypes. Such comparative analyses could clarify whether the seven-cluster solution observed in Japan reflects uniquely Japanese configurations or represents one instantiation of a broader, culturally moderated typological structure. This approach would directly address concerns about Western-centric bias in personality theory and contribute to a more globally inclusive model of personality organization.

Finally, future studies should integrate longitudinal and clinically relevant outcome measures to examine the developmental and adaptive significance of personality prototypes. Longitudinal designs would enable researchers to assess the stability and change of prototypes across time, as well as their predictive validity for mental health, well-being, and psychosocial functioning. Incorporating indicators such as anxiety, depression, stress responses, or treatment outcomes would allow for direct evaluation of the clinical utility of HEXACO-based prototypes. Such extensions would move person-centered personality research beyond descriptive classification toward a more explanatory and applied science of psychological adaptation.

VI. CONCLUSION

~~This study identified seven distinct personality types in a sample of Japanese individuals based on the HEXACO model. These clusters offer new insights into the structure of Japanese personalities, while also reflecting both universal and culturally specific aspects of personality. Particularly notable was the influence of the Honesty–Humility factor, which plays a significant role in shaping personality types within the context of Japanese social norms, as well as the relatively lower levels of Extraversion and Agreeableness observed in this sample. These findings highlight the importance of cultural factors in personality formation and provide a valuable foundation for future research into personality types in non-Western contexts. The identified personality types can also be applied in fields such as clinical psychology, education, and organizational management, helping to develop tailored support systems and management strategies that align with the unique personality profiles observed in Japanese individuals. Moving forward, further exploration of cross-cultural differences and similarities in personality types, as well as an examination of the relationships between personality traits and psychological outcomes such as well-being and risk-taking behavior, will enrich our understanding of personality in a globalized world. This study serves as a step toward such comprehensive research, with implications for both theory and practice.~~

This study provides robust evidence for culturally grounded personality prototypes within the Japanese population using the HEXACO framework. By applying a multi-method clustering strategy to a large non-WEIRD sample, seven distinct and replicable personality types were identified, demonstrating that personality structure at the person-centered level reflects both universal trait dimensions and culturally specific patterns of covariation. These findings extend previous Western-centered research by showing that greater typological differentiation may emerge when personality is examined within its sociocultural context.

A central contribution of the present work lies in highlighting the theoretical and empirical importance of the Honesty–Humility dimension in organizing personality types. Beyond its established predictive value at the trait level, Honesty–Humility played a key role in distinguishing cooperative, reliable profiles from more self-oriented and emotionally volatile configurations. This underscores the value of the HEXACO model for capturing moral and interpersonal aspects of personality that are especially salient in culturally embedded social systems.

Overall, the findings advance cross-cultural personality psychology by integrating person-centered methodology, large-scale data, and culturally informed interpretation. The present study demonstrates that personality types are not merely simplified representations of traits, but meaningful

configurations shaped by cultural norms, social expectations, and psychological adaptation. By providing a rigorous and culturally sensitive model of personality structure in Japan, this research contributes to the development of a more globally inclusive and theoretically integrated personality science.

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Prior to conducting the study, I consulted with the Research and Development Division of the Health Policy Bureau at the Ministry of Health, Labour and Welfare. It was confirmed that ethical approval was not required for this study in accordance with local legislation and institutional requirements, specifically Chapter 1, Part 3, Section C of the Ethical Guidelines for Medical and Biological Research Involving Human Subjects issued by the Ministry of Health, Labour and Welfare, Japan. After explaining the research content, the Ministry further confirmed that this study does not raise ethical concerns under the above-mentioned guidelines. Accordingly, all procedures were conducted in compliance with these national ethical standards, including informed consent, confidentiality, and the protection of participants' rights.

Informed consent was obtained from all participants, with clear information provided on the study's purpose, data collection methods, privacy protection measures, voluntary nature of participation, and the option to withdraw at any time. To protect participant privacy, data were collected anonymously, securely stored, and participants were informed of their right to request correction or deletion of their data. Informed consent for the publication of the research results was obtained from all participants involved in this study.
Consent to publish the research findings was obtained from all participants at the time of informed consent. Participants were informed that anonymized and aggregated results may be published in academic journals.

The data that support the findings of this study are available in Research Square at https://doi.org/10.21203/rs.3.rs-5323735/v1, including the datasets 4f598e5790e022899ae145f3.csv and 18972bcadd819eb918f5698c.csv.

The author developed the concept, designed the study, collected and analyzed the data, and wrote the manuscript. The author read and approved the final manuscript.

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